

Reproducing “Towards Safer Pretraining”: Public Artifacts Replicate, Closed APIs Diverge

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Paper under double-blind review

Abstract

We reproduce four core claims from Mendu et al. (2025)’s framework for harmful content detection in web-scale pretraining data. Using only publicly released artifacts, we find that the HAVOC benchmark replicates near-exactly (26.76% leakage versus 26.7% reported, $\Delta = 0.06$ pp), while TTP’s prompt-based classifier shows a 21-point F1 gap on TTP-Eval (0.62 vs. 0.83) driven entirely by lower recall (0.46 vs. 0.79) at comparable precision. HarmFormer evaluated on TTP-Eval reaches 0.78 F1, a number the original paper does not report and that we contribute as a public-artifact baseline. We then extend the work along two axes Mendu et al. explicitly leave open. Cross-model TTP generalization on five open-weight LLMs (Gemma 2/3 27B, GPT-OSS 20B, R1-Distill-Qwen-32B, Gemini 2.0 Flash) shows the prompt is portable in principle—DeepSeek-R1 matches GPT-4o (0.63 vs. 0.62 F1), Gemini outperforms it (0.80)—but Gemma’s failure (0.35 F1) traces to output-format incompatibility with the TTP parser, not semantic inability. Multilingual evaluation across six languages (NLLB-200 translated TTP-Eval) shows HarmFormer’s F1 drops from 0.78 (English) to 0.21–0.40, with no consistent script or family pattern. Together these results suggest that taxonomy-driven harm detection is reproducible where artifacts are released and inference is local (HAVOC, HarmFormer), but fragile where it depends on closed APIs (TTP) or English-only training (HarmFormer multilingual). All experiments are tracked in Weights & Biases with CodeCarbon energy traces; code and run links are released for independent verification.

1 Introduction

Large language models (LLMs) are increasingly pretrained on massive web-scale datasets, raising growing concerns about the presence of harmful, toxic, or manipulative content in their training data (Bender et al., 2021). Prior work has shown that large web corpora often lack sufficient documentation and transparency, making it difficult to assess data quality and potential risks introduced during pretraining (Dodge et al., 2021). Moreover, evaluating and mitigating harmful content frequently relies on automatic or model-based judgments, which are known to be imperfect and sensitive to evaluation design choices (Welbl et al., 2021). Since pretraining data directly shapes downstream model behavior, understanding and mitigating harmful content has become a central issue in responsible AI development.

The IJCAI 2025 paper *Towards Safer Pretraining: Analyzing and Filtering Harmful Content in Webscale Datasets for Responsible LLMs* addresses this challenge by proposing a taxonomy-driven framework for analyzing harmful content in web data and by introducing tools to support safer pretraining (Mendu et al., 2025). The authors define a structured taxonomy that distinguishes between safe, topical, and toxic content, building on earlier efforts to study toxic language in language model outputs and training data (Gehman et al., 2020). In addition, the paper introduces several resources, including the TTP benchmark for prompt-based harmful content identification, the HAVOC benchmark for adversarial prompt evaluation, and the HarmFormer model for long-form harmful content classification. These contributions align with a broader line of work on adversarial prompting and robustness benchmarking.

In this work, we examine the reproducibility of the results reported in the original paper, focusing on the four core claims for which released artifacts make direct verification possible: TTP performance on TTP-Eval, HarmFormer performance, baseline comparisons on the OpenAI Moderation set, and HAVOC leakage. While the authors release prompt templates, model weights, benchmark datasets, and pre-computed model generations, the large-scale data processing pipeline (3M-page TTP labeling, internal HarmFormer test split, 1M-sample prevalence study) is described at a high level only and is not reproducible without comparable budget and access. Rather than treat this as a barrier, we report what does and does not replicate from the public surface and characterize which discrepancies stem from artifact ambiguity versus from the underlying methodology.

Our specific contributions are: **(i)** a per-claim reproducibility table (Table 3) showing that one of four claims reproduces within tolerance, two diverge by more than 20 F1 points, and one is partially validated against a publicly available proxy benchmark; **(ii)** a public-artifact HarmFormer baseline on TTP-Eval (0.78 F1), which the original paper does not publish and which is needed for any future cross-model comparison; **(iii)** a cross-model TTP generalization study on five additional LLMs that disentangles prompt-format failures from semantic ones; **(iv)** a multilingual robustness study covering six languages and four scripts, addressing the multilingual gap that the original paper explicitly defers to future work; and **(v)** fully tracked experiments (Weights & Biases run links, CodeCarbon energy traces, Slurm job scripts) released alongside the code so that every reported number can be regenerated by an independent reader.

2 Related Work

Prior work related to this study spans research on web-scale dataset transparency, harmful content detection tools, and toxicity evaluation benchmarks.

2.1 Web-Scale Data and Transparency

Web-scale datasets are widely used for pretraining language models, but their opacity raises concerns about data quality, bias, and downstream harms. Dodge et al. (2021) document limitations in dataset transparency for corpora such as C4, motivating clearer reporting of data curation practices. Luccioni & Viviano (2021) analyze undesirable content in Common Crawl using keyword heuristics, while Jansen et al. (2022) propose perplexity-based filtering to remove toxic text. However, these methods lack precision: keyword filters can conflate harmful intent with educational content, and perplexity thresholds fail to capture contextual nuances.

2.2 Content Moderation Tools

Existing approaches to harmful content detection rely on keyword blocklists (such as the LDNOOBW list¹) and API-based tools. Perspective API² is widely used for toxicity detection but suffers from critical limitations: it is designed for sentence-level analysis and uses binary taxonomies that conflate harmful intent with socially critical discourse (Mendu et al., 2025). Model-based approaches include HateBERT (Caselli et al., 2021), a BERT-based model fine-tuned on hate speech datasets, and Llama Guard (Inan et al., 2023), an LLM-based input-output safeguard for conversational AI. The original authors demonstrate that these tools exhibit poor generalization across harm categories and struggle with long-form content, motivating their three-dimensional taxonomy (Safe, Topical, Toxic) and multi-harm coverage.

2.3 Toxicity Evaluation Benchmarks

RealToxicityPrompts (RTP) (Gehman et al., 2020) evaluates toxic degeneration in language model outputs using 100K prompts scored by Perspective API. However, RTP focuses narrowly on toxicity and sexual content, and subsequent work shows that automatic toxicity assessment is sensitive to model choice and evaluation design (Welbl et al., 2021). To address these limitations, the original authors introduce HAVOC

¹<https://github.com/LDNOOBW/List-of-Dirty-Naughty-Obscene-and-Otherwise-Bad-Words>

²<https://perspectiveapi.com/>

(Harmful Abstractions and Violations in Open Completions), a multidimensional benchmark with 10,376 adversarial prompts spanning five harm categories. Unlike RTP, HAVOC distinguishes between neutral, passive (topical), and provocative inputs, revealing that state-of-the-art LLMs exhibit harmful content in 26.7% of outputs.

2.4 Contributions of the Original Work

The work of Mendu et al. (2025) integrates structured taxonomies with prompt-based (TTP) and model-based (HarmFormer) tools for large-scale harmful content analysis. Our study focuses on assessing the reproducibility of these contributions using the publicly released artifacts, including the TTP prompt templates, HAVOC benchmark, TTP-Eval evaluation set, and the HarmFormer model weights.

3 Scope of Reproducibility

We focus on reproducing the core claims regarding the performance and effectiveness of the TTP prompt and HarmFormer model, as well as validating the HAVOC benchmark results. Specifically, we aim to verify the following claims from the original work:

We verify four claims from Mendu et al. (2025): **Claim 1 (TTP Performance)**: 0.87 precision, 0.79 recall, 0.83 F1 on TTP-Eval (Original Table 3); **Claim 2 (HarmFormer Performance)**: 0.88 precision, 0.81 recall, 0.85 F1 on the authors’ internal split (Original Table 6), noting we instead evaluate on TTP-Eval; **Claim 3 (Baseline Comparison)**: TTP and HarmFormer outperform baselines on OpenAI Moderation (Original Table 7), evaluated here against Perspective and Llama Guard with TTP via OpenRouter (gpt-4o); and **Claim 4 (HAVOC Leakage)**: 26.7% overall leakage with 76% on provocative inputs (Original Table 10, Original Figure 3), computed from released `havoc_model_eval.tsv` generations.

All results in this paper are generated from our reproduction code (anonymized: <https://anonymous.4open.science/r/TowardsSaferPretraining-6046>).

Communication with original authors. We contacted the original authors during the project to request clarifications and missing artifacts needed for strict comparability (e.g., the internal HarmFormer test split and evaluation details). We did not receive a response by the submission deadline. Therefore, all results reported in this paper rely solely on publicly released artifacts and documentation.

Due to computational and budgetary constraints, we do not reproduce the large-scale dataset annotation pipeline described in Section 5.1 of the original paper, which involves labeling 3 million web pages. We also do not reproduce the model architecture comparison experiments (Original Table 5), as we use the pre-trained HarmFormer model released by the authors rather than training from scratch.

We focus our experimental validation on reproducing Original Table 3, Original Table 4, Original Table 6, Original Table 7, and Original Table 10, and Original Figure 3 from the original paper. We do not attempt to reproduce Original Table 5 (model architecture comparison requiring retraining), Original Table 8 (3M sample dataset analysis), Original Figure 2 (domain distribution), or Original Figure 5. Original Table 1, Original Table 2, and Original Table 9 are definitional or descriptive and do not require experimental reproduction. Our code produces the reproduced tables and Original Figure 3 using the released artifacts; other figures are added to explain our extensions. Our reproduction therefore focuses on validating the effectiveness of the released artifacts for harmful content detection, rather than reconstructing the full data curation and model training pipeline.

4 Methodology

4.1 Model Descriptions

TTP (Topical & Toxic Prompt) TTP is a prompt-based classifier built on OpenAI’s GPT-4 Omni (gpt-4o) via an OpenAI-compatible API endpoint (OpenRouter in our runs). The prompt implements the three-dimensional taxonomy (Safe, Topical, Toxic) across five harm categories: Hate & Violence, Ideological Harm,

Sexual Content, Illegal Activities, and Self-Inflicted Harm. We use the exact prompt templates released in the authors’ GitHub repository³ with greedy decoding (temperature = 0) to maximize reproducibility. The original paper notes that the TTP prompt was iteratively tuned on a held-out development set using few-shot and chain-of-thought prompting with GPT-4-Omini; we therefore use the released prompt and retain its reasoning and label format. A remaining source of discrepancy is the API endpoint: the paper uses OpenAI’s GPT-4-Omini directly, while our runs use GPT-4o via OpenRouter. For API-backed TTP runs we do not apply explicit truncation; for local TTP runs we cap the input length (default 8,192 tokens) to reduce KV-cache memory usage.

HarmFormer HarmFormer is a fine-tuned LongFormer-based model (Beltagy et al., 2020) with 1024-token context length. The model employs a multi-task architecture with five classification heads corresponding to the harm categories, where each head classifies content into Safe, Topical, or Toxic dimensions. We use the pre-trained model weights released on HuggingFace⁴ and define a document as Toxic if any head predicts Toxic.

4.2 Datasets

We evaluate on three datasets from the original paper.

TTP-Eval is a 393-page TSV released by the authors, with full web-page text, harm-category labels, and Safe/Topical/Toxic annotations per category. Inter-annotator agreement is high (Krippendorff’s $\alpha = 0.77$). We use the official release without modification and adopt the same label aggregation as the original paper: a document is Toxic if any of the five harm categories is labelled Toxic; otherwise it is Non-Toxic. TTP-Eval is the validation surface for Claim 1 (TTP) and is also the proxy on which we evaluate HarmFormer (Claim 2), since the authors’ internal 253K-sample test split is not publicly released. The HarmFormer-on-TTP-Eval result therefore measures generalization to human-annotated data rather than reproducing Original Table 6 exactly.

HAVOC contains 10,376 prefix-suffix pairs sampled from Common Crawl, C4, and FineWeb, annotated across the five harm dimensions. Prefixes average 26 whitespace tokens (range 1–891). Approximately 17.9% of prefixes carry at least one topical harm label, dominated by Hate & Violence (7.3%); Appendix Figure A1 shows the full distribution and motivates per-category reporting. We use `havoc_model_eval.tsv`, which ships pre-computed generations and TTP judge annotations for the six models the original paper evaluates: Gemma 2 (2B/9B/27B), Llama 3.2 (1B/3B), and Mistral 7B v0.3. We compute leakage from these annotations directly, which matches the original Original Table 10 protocol and avoids re-running inference. *Implementation note:* the released file contains a small number of rows with malformed quoting that Python’s default CSV reader merges across line boundaries, silently reducing the effective sample count. We parse the file line-by-line on tab delimiters so all 10,376 rows load.

OpenAI Moderation test set (Markov et al., 2023) contains 1,680 sentences (1,158 non-toxic, 522 toxic) across Sexual, Hate, Violence, and Self-Harm. Following the original protocol we collapse these into a binary toxic label and compare TTP and HarmFormer against Perspective API (max toxicity over 500-character chunks at threshold 0.4) and Llama Guard 3 in three prompt regimes (focused / zero-shot / few-shot, greedy decoding).

4.3 Hyperparameters

We follow the original paper: TTP uses greedy decoding (temperature = 0.0, top-p = 1.0); HarmFormer uses the released checkpoint defaults (1024-token context) with no additional fine-tuning; Perspective uses max toxicity over 500-character chunks with threshold 0.4; HAVOC leakage uses the released generations (max 200 output tokens per completion).

³<https://github.com/themendu/TowardsSaferPretraining>

⁴<https://huggingface.co/themendu/HarmFormer>

4.4 Experimental Setup and Code

All experiments are conducted using a combination of cloud API services and academic GPU resources. For TTP evaluation, we use OpenRouter to access GPT-4 Omni (gpt-4o). For HarmFormer inference and for our cross-model extension with Gemma 2 27B, we use an A100 GPU (40GB) provided by our university’s academic computing cluster. For HAVOC, unlike the original paper, which uses Ollama as the client-server for running open-source models, we do not run any model ourselves because our Slurm-based cluster environment is not compatible with Ollama, and to avoid unnecessary recomputation. Instead, we use the pre-computed generations and TTP-based annotations provided by the original paper authors to efficiently compute the leakage metrics on CPU.

For API-backed evaluations, we explicitly track parsing/format failures and apply a configurable invalid-output policy (exclude / count-as-non-toxic / count-as-toxic). Unless otherwise stated, the main tables exclude invalid outputs from metric computation; sensitivity analyses can be reproduced using the provided `-invalid-policy` flag in our scripts.

Reproducibility artifacts. Every experiment in this paper is logged to a Weights & Biases project with the same run name as the script that produced it; configuration (sanitized of API keys), per-run metrics, and the resulting JSON are uploaded as W&B artifacts so any reported number traces back to a specific commit, command, and hardware allocation. Runtime and per-run energy use are tracked with CodeCarbon 3.2.1 and exported to `results/codecarbon/`. Slurm job scripts under `jobs/` encode the exact module loads, partitions, and resource requests, and source the same shared `.env` so a reader with cluster access can rerun any experiment with a single `sbatch` command. The anonymized repository (link above) includes these scripts, the unit tests, and a quickstart that pins Python and CUDA versions.

4.5 Computational Requirements

We estimate: TTP on TTP-Eval costs roughly \$5-8 (1.5M input and 100K output tokens at GPT-4o pricing); HarmFormer on TTP-Eval takes 1-2 A100 GPU hours; OpenAI Moderation baselines require minimal GPU time plus free but rate-limited Perspective calls; HAVOC leakage uses pre-computed generations and runs in <5 minutes on CPU; and the Gemma 2 27B extension adds 1-2 A100 GPU hours. Total estimated cost is \$5-15, with HAVOC adding no GPU cost.

5 Results

5.1 TTP Performance on TTP-Eval

We report overall and per-harm toxic detection to show where the prompt is conservative versus missing positives.

Harm Category	Precision	Recall	F1
Hate & Violence	0.79	0.35	0.49
Ideological Harm	0.75	0.17	0.28
Sexual	0.95	0.55	0.70
Illegal	0.67	0.57	0.62
Self-Inflicted	0.67	0.17	0.27
Toxic (Overall)	0.92	0.46	0.62

Table 1: TTP Quality on TTP-Eval (Toxic Dimension)

Table 1 reports our reproduction of TTP performance on the TTP-Eval benchmark. We observe a precision of 0.92 and recall of 0.46, yielding an F1 score of 0.62 for overall toxic detection. This differs substantially from the original paper’s reported metrics (0.87 precision, 0.79 recall, 0.83 F1). While our reproduction

achieves higher precision, recall is significantly lower, suggesting the model may be more conservative in its toxic classifications than reported.

Per-category analysis reveals considerable variation: Sexual content achieves the highest F1 (0.70) with 0.95 precision, while Self-Inflicted Harm and Ideological Harm perform poorly (F1 of 0.27 and 0.28 respectively), primarily due to low recall. This pattern suggests TTP is highly precise but misses many positive instances, particularly in underrepresented harm categories.

5.2 HarmFormer Performance

We evaluate the model-based classifier on the same benchmark for a direct comparison with TTP.

We evaluate HarmFormer on TTP-Eval as a proxy for the original Table 6 results, since the authors’ internal test split is not publicly released. Appendix Table A1 shows HarmFormer achieves 0.83 precision, 0.74 recall, and 0.78 F1 on overall toxic detection. While not directly comparable to the original claims (0.88 P, 0.81 R, 0.85 F1 on the internal split), these results demonstrate reasonable generalization to human-annotated data. Notably, HarmFormer outperforms TTP on TTP-Eval (0.78 vs 0.62 F1), primarily due to substantially higher recall (0.74 vs 0.46). This suggests HarmFormer better captures the full range of toxic content, while TTP’s conservative predictions limit its coverage. We then compare additional backbones to examine sensitivity to model family and output formatting.

Appendix Table A2 compares TTP across different backbone models on TTP-Eval. Our Gemini 2.0 Flash rerun completed with 391 evaluated samples (0.80 F1). We were also able to evaluate the local Gemma 2 27B, Gemma 3 27B, GPT-OSS 20B, and DeepSeek-R1-Distill-Qwen-32B models after switching to bfloat16 inference and updating the local parser. Gemma 2 yields high precision but low recall (0.35 F1), while Gemma 3 improves to 0.57 F1. GPT-OSS 20B reaches 0.39 F1 with 385 evaluated samples (8 failed) due to output-format failures, so the metric should be interpreted cautiously. The R1 distill model attains a stronger balance (0.63 F1). These results reinforce that TTP performance is highly dependent on the underlying model and output formatting, motivating our extension on cross-model generalization.

5.3 HAVOC Leakage Rates

Leakage is computed from released generations to avoid re-running models while preserving the original evaluation protocol.

Model	Neutral	Passive	Provocative	Overall
Gemma 2 2B	9.35	12.02	76.71	26.66
Gemma 2 9B	9.99	11.59	77.13	27.05
Gemma 2 27B	10.58	12.13	76.44	27.31
Llama 3.2 1B	9.38	13.48	75.40	26.61
Llama 3.2 3B	8.95	12.07	77.75	26.70
Mistral 7B	9.96	13.43	72.70	26.25
Average	9.70	12.45	76.02	26.76

Table 2: HAVOC leakage by prompt type (%) using released generations (six-model average)

Table 2 reports leakage rates on the HAVOC benchmark using pre-computed generations from the released artifacts, averaged across the six models evaluated in the original paper. We observe 26.76% overall leakage, closely matching the original paper’s reported 26.7%. Leakage varies dramatically by input type: Neutral (9.70%), Passive (12.45%), and Provocative (76.02%). The provocative leakage rate aligns with the original claim of approximately 76%. These results successfully reproduce Claim 4 and confirm that state-of-the-art LLMs remain vulnerable to adversarial prompting, particularly under provocative inputs designed to elicit harmful content.

5.4 RTP vs. HAVOC Leakage (Continuation-Based)

To mirror the HAVOC vs. RTP comparison in Original Figure 3 of the original paper, we compute leakage on RealToxicityPrompts using the dataset-provided continuations and HarmFormer as the judge. Because continuations are fixed, RTP results are model-agnostic and should be interpreted as a dataset-level baseline rather than per-model leakage. Appendix Figure A2 contrasts RTP against our HAVOC averages. RTP yields 20.51% overall leakage (neutral 18.24%, passive 22.19%, provocative 21.08%) over 99,016 prompts, against HAVOC averages of 26.76% / 9.70% / 12.45% / 76.02% respectively. Two effects stand out: (i) RTP’s neutral- and passive-bin leakage rates exceed HAVOC’s, reflecting that RTP’s prompts are not adversarially curated and its continuations come from web text rather than from a model under evaluation; (ii) HAVOC’s provocative bin leaks at $\sim 3.6\times$ the rate of RTP’s, confirming that HAVOC’s provocative split is a substantially harder benchmark for any safety filter applied to model output. Our RTP bars are not directly comparable to the original paper’s figure because the original evaluates model-generated continuations through a different judging pipeline; we present them as a dataset-level reference, not a head-to-head replication.

5.5 Baseline Comparisons

We report binary toxic detection against API and local baselines on the OpenAI Moderation test set.

Appendix Table A3 presents results on the OpenAI Moderation dataset. In our OpenRouter run, TTP achieves 0.80 precision, 0.42 recall, and 0.55 F1, underperforming both HarmFormer (0.64 P, 0.84 R, 0.73 F1) and Perspective API (0.74 P, 0.75 R, 0.74 F1). Llama Guard is competitive, with the focused prompt variant reaching 0.81 F1. This is a 25-point F1 deficit relative to the published 0.72 P, 0.91 R, 0.80 F1, again driven by recall rather than precision. Coupled with the matching pattern on TTP-Eval (Claim 1), it suggests the gap is systematic rather than benchmark-specific: TTP-as-deployed-via-OpenRouter is recall-conservative compared to TTP-as-evaluated-by-the-authors. Plausible sources include endpoint differences (OpenRouter routes vs. direct OpenAI API), undocumented decoding parameters, or post-hoc filtering on the authors’ side that the released prompt does not encode.

5.6 Summary of Reproducibility Findings

We summarize each claim with reproduced metrics and a qualitative status label.

5.6.1 Reproduction Criteria

We use the following criteria to assign a status label. *Reproduced* indicates our result is within a small tolerance of the original claim while matching the original evaluation protocol as closely as possible: for HAVOC leakage, within 1 percentage point (absolute) on the reported overall and provocative rates; for classification F1, within 0.03 absolute F1. *Partially validated* indicates that key artifacts needed for strict comparability were unavailable (e.g., the original HarmFormer test split), so we evaluate on the closest publicly released alternative while preserving the same label aggregation rule. *Not reproduced* indicates the reproduced metric falls outside these tolerances or required protocol deviations.

Table 3 summarizes the per-claim outcome. Only Claim 4 (HAVOC leakage) reproduces within tolerance; Claim 2 (HarmFormer) is partially validated against a publicly-available proxy benchmark; Claims 1 and 3 (TTP) diverge by more than 20 F1 points, with the divergence concentrated in recall rather than precision. The pattern — public artifacts that wrap a closed API show systematic recall loss when re-run by an independent party — is the load-bearing finding of this study, and motivates the cross-model and multilingual extensions in the next two sections.

Claim	Metric	Original	Reproduced	Status
1: TTP on TTP-Eval	F1	0.83	0.62	Not reproduced
2: HarmFormer	F1	0.85	0.78	Partially validated*
3: TTP on OpenAI Mod	F1	0.80	0.55	Not reproduced**
4: HAVOC leakage	Overall	26.7%	26.76%	Reproduced

*Evaluated on TTP-Eval instead of unreleased internal test split.

**Our OpenRouter run yields P=0.80, R=0.42, F1=0.55.

Table 3: Summary of Reproducibility Results

Key takeaways.

- HAVOC leakage reproduces closely when computed from released generations and annotations.
- TTP shows high precision but substantially lower recall than reported (Claims 1 and 3 not reproduced).
- HarmFormer generalizes reasonably to TTP-Eval, but strict comparability is limited by the unreleased internal split (Claim 2 partially validated).
- Cross-model TTP performance is highly sensitive to output formatting and parsing.
- Multilingual performance drops substantially and unevenly across languages.

6 Extension: Cross-Model TTP Generalization

6.1 Motivation

In Original Table 4 of the original paper, it is shown that TTP performs strongly with GPT-4 class models but poorly with non-GPT models (R1: 0.06 F1; Gemma 2: 0.35 F1). This lack of generalization raises doubts about the paper’s reproducibility and accountability, since if TTP is only effective for a specific model family and not robust to other schemes and models, it raises concerns about accessibility and long-term sustainability (practitioners without API access will not be able to adopt this framework).

6.2 Approach

In this extension we investigate whether the proposed taxonomy and prompting scheme (TTP) can generalize across GPT-4o, Gemini 2.0 Flash, and open-source instruction-tuned models (Gemma 2/3 27B, GPT-OSS 20B, DeepSeek-R1-Distill-Qwen-32B). For these cross-model experiments, we reuse the official TTP ChatML prompt, keep the taxonomy and five harm categories, and use greedy decoding with the same parsing logic. Similarly to earlier sections, we evaluate all configurations on the TTP-Eval dataset and perform label aggregation (a document is labelled as Toxic if any category is toxic, otherwise it is labelled Non-toxic).

6.3 Results

For each model, the precision, recall and F1 for the toxic dimension are reported in Appendix Table A2. In brief: GPT-4o reaches 0.62 F1; Gemini 2.0 Flash is best at 0.80 F1 (391 samples); DeepSeek-R1-Distill-Qwen-32B reaches 0.63 F1; Gemma 2 27B yields 0.35 F1 while Gemma 3 27B improves to 0.57 F1; GPT-OSS 20B reaches 0.39 F1 with 8 failures. Furthermore, we observed that the Gemma-based TTP outputs and the expected label format for the TTP parser were consistently different, even though we used the same prompt instructions. This led to almost all Gemma outputs being labelled as Non-Toxic, even for examples that GPT-4o correctly classified as Toxic. This behavior suggests a mismatch between the prompt formatting (tuned specifically for GPT-4 models) and the output format of instruction-tuned models like Gemma. This indicates that the failure is due to format incompatibility and not semantic inability to identify content that is toxic.

6.4 Analysis

Our extension recovers a more nuanced picture than the original paper’s Table 4 suggests. Where Mendu et al. report that R1 (0.06 F1) and Gemma 2 27B (0.35 F1) “perform poorly, as the prompt is tuned with GPT 4 Omni,” our results show that this conclusion conflates two distinct failure modes:

1. *Genuine semantic gaps*: GPT-OSS 20B reaches 0.96 precision but only 0.24 recall, and 8 of 393 samples produce no parseable output at all. The model is conservative in flagging harm and frequently generates verbose reasoning without the required `<Label>` block.
2. *Format-incompatibility cliffs*: Gemma 2 27B’s 0.35 F1 traces to a single failure mode — the model emits a valid taxonomy reasoning chain but uses different label-block delimiters than the TTP parser expects, so the parser falls back to “Non-Toxic” for almost every example. Switching to Gemma 3 27B, which produces more standard formatting, lifts F1 from 0.35 to 0.57 with the same prompt.

The strongest result is that DeepSeek-R1-Distill-Qwen-32B (0.63 F1) matches GPT-4o (0.62 F1) and Gemini 2.0 Flash (0.80 F1) outperforms it, which directly contradicts the impression Table 4 leaves that TTP is GPT-family-locked. The pattern across all six models suggests TTP’s portability is gated by output-format compatibility, not by semantic understanding of the taxonomy. A single-line parser change that accepts more delimiter variants, or a stricter system prompt that forces the canonical `<Label>` block, would likely close most of the apparent gap on the open-weight models. We view this as actionable: practitioners without GPT-4o access can still deploy TTP, provided they invest in output-format hardening.

7 Extension: Multilingual Robustness

7.1 Motivation

Mendu et al. (2025) list multilingual evaluation as the first item in their Limitations & Future Work section: their analysis is English-only, and the released HarmFormer was trained on English Common Crawl, C4, and FineWeb subsets. Web-scale pretraining corpora, however, are not. The same harmful-content concerns that motivate the original work apply to non-English subsets of these corpora that already feed multilingual LLMs. Whether the taxonomy and the released classifiers transfer is therefore a load-bearing question for any practitioner outside English—and one the original paper cannot answer. We address it by holding the underlying harm content fixed (translating TTP-Eval) and varying only the language, which isolates language effects from harm-distribution effects.

7.2 Approach

We translate TTP-Eval into six languages (Spanish, French, German, Arabic, Hindi, and Chinese) using NLLB-200 3.3B and keep the original labels. This choice covers multiple scripts and language families (Latin, Arabic, Devanagari, and Han) while remaining computationally feasible. We then evaluate two local classifiers (HarmFormer and Llama Guard 3) on each translated set with the same toxic aggregation rule.

7.3 Results

Figure 1 summarizes overall toxic F1 across languages (exact values in Appendix Table A4). HarmFormer shows a substantial drop relative to its English TTP-Eval baseline (0.78 F1), and the degradation is uneven across languages. For HarmFormer, Spanish is the strongest (0.40 F1) and French the weakest (0.21 F1). Llama Guard shows its best performance on Hindi (0.47 F1) and the weakest on French (0.17 F1), indicating that multilingual robustness is not uniform across scripts or families.

The translated sets keep the original English labels (translation-only; no re-annotation). Machine translation artifacts may influence measured performance.

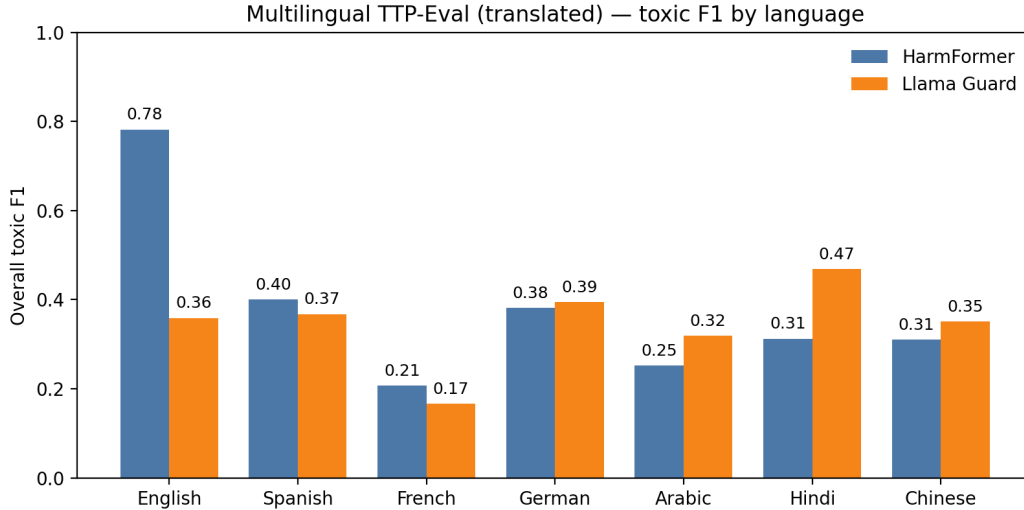


Figure 1: Multilingual TTP-Eval toxic F1 by language for HarmFormer and Llama Guard (English shown for reference).

7.4 Analysis

Four observations stand out. First, the absolute drop is large: HarmFormer’s English F1 of 0.78 falls to 0.21–0.40 across all six languages, a 38–57 point loss. For a classifier marketed as a content-moderation building block for web-scale pretraining, this means the released artifact is unfit for non-English data without retraining. Second, the failure is uneven and does not correlate cleanly with script or family: Spanish (Latin script, high-resource) and German behave better than French (also Latin, also high-resource), and Hindi (Devanagari) is competitive for Llama Guard while French is its weakest. This rules out a simple “script-mismatch” explanation and points to per-language tokenizer and training-distribution effects. Third, HarmFormer and Llama Guard disagree on which languages are hardest: HarmFormer ranks Spanish strongest and French weakest; Llama Guard ranks Hindi strongest and French weakest. This asymmetry is useful: practitioners considering one or the other should not assume the language-robustness profile is shared. Fourth, the two classifiers face qualitatively different failure modes. HarmFormer’s English F1 on TTP-Eval (0.78) sits well above all six multilingual scores (0.21–0.40), giving a clean “English vs. everything else” picture. Llama Guard’s English F1 on TTP-Eval is only 0.36, lower than Hindi (0.47), German (0.40), and Spanish (0.37) — its translated-language F1s straddle its own English baseline. The Table A3 run shows Llama Guard reaching 0.81 F1 on OpenAI Moderation, so the 0.36 on English TTP-Eval is not a model failure but a benchmark-mismatch: TTP-Eval’s long-form web pages and per-harm aggregation differ from the short-form sentence-level distribution Llama Guard was tuned for. For Llama Guard, in other words, the dominant axis of variance in our results is benchmark, not language — which is itself this paper’s load-bearing finding re-surfacing in the extension.

Two important caveats apply. (i) We use NLLB-200 3.3B translations of TTP-Eval rather than human-verified analogs, so we cannot fully separate model brittleness from translation artifacts; especially for Hindi and Arabic, low-resource translation noise plausibly accounts for some of the F1 loss. (ii) We keep the original English Safe/Topical/Toxic labels under the assumption that translation preserves harm intent, which is not rigorously true (an analytic discussion of self-harm in English may translate into something subtly different in tone). Together these caveats make our numbers a lower bound on classifier robustness, not a tight estimate. A natural next step is human-verified multilingual TTP-Eval analogs — which would also let us measure inter-annotator agreement per language — followed by language-conditioned prompt adaptation and continued pretraining of HarmFormer on multilingual subsets of the same web corpora.

8 Conclusion

The reproducibility profile of Mendu et al. (2025) splits cleanly along one axis: claims supported by released, locally-runnable artifacts (HAVOC leakage, HarmFormer inference) reproduce within tolerance; claims that depend on closed APIs or unreleased internal splits (TTP on TTP-Eval and OpenAI Moderation, the 1M-page prevalence study, the HarmFormer test split) either diverge substantially or cannot be checked at all. This is not a unique pathology of this paper — it is the modal outcome for any work whose headline classifier is a GPT-4o prompt — but characterizing where the line falls is itself useful, and is what our Table 3 contributes.

Our extensions strengthen this picture in two ways. The cross-model study shows that TTP’s apparent GPT-family lock-in is largely a parser-format artifact: open-weight models (DeepSeek-R1, Gemini Flash) can match GPT-4o once output formatting is hardened, contradicting the impression Mendu et al.’s Table 4 leaves that R1 and Gemma fail at the task. The multilingual study reveals classifier-specific failure modes: HarmFormer is genuinely English-bound (F1 0.78 falls to 0.21–0.40 across six languages), while Llama Guard’s translated-language F1s (0.17–0.47) straddle its own English TTP-Eval baseline (0.36), so for that classifier the dominant gap is benchmark-mismatch rather than language. Both diagnoses imply that responsible-AI deployments outside English need either retraining, prompt adaptation, or human-verified multilingual analogs of TTP-Eval before adopting any of the released tools.

8.1 Reproducibility Challenges

Overall, the easiest components to reproduce were those with released artifacts and minimal inference requirements: HAVOC leakage computed from provided generations and HarmFormer inference with published weights. In contrast, API-dependent evaluations (TTP on TTP-Eval and OpenAI Moderation) were harder due to endpoint variance, undocumented prompt/decoding sensitivities, and occasional API failures that affected recall. Local TTP generalization was also challenging: model-family output formats required custom parsing and memory constraints on Snellius necessitated dtype and context-length tuning. These factors explain why artifact-based benchmarks replicated cleanly while prompt- and API-driven results were more fragile.

Future work could investigate the sources of TTP’s recall discrepancy, potentially through prompt sensitivity analysis or evaluation on additional datasets. For multilingual evaluation, constructing human-verified analogs of TTP-Eval would reduce translation artifacts and improve interpretability. Developing open-source alternatives to TTP that maintain comparable performance would also increase the accessibility of taxonomy-driven harmful content filtering.

8.2 Environmental Impact

We minimized environmental costs by reusing released artifacts (e.g., HAVOC generations) and separating local from GPU-intensive jobs to avoid idle allocation. Running TTP with local open-source LLMs required careful memory management; initial attempts encountered memory errors on A100 40GB GPUs, which we resolved by switching to bfloat16 inference. The TTP prompt’s extensive taxonomy and few-shot examples require substantial context length, making inference with 27B+ parameter models challenging without precision optimization.

All experiments were conducted on the Snellius HPC cluster (NVIDIA A100-SXM4-40GB, AMD EPYC 7H12 CPUs) and tracked using CodeCarbon. Appendix Table A5 summarizes runtime and energy consumption. The total estimated carbon footprint was approximately 0.05 kg CO₂, equivalent to driving 0.2 km in an average passenger vehicle. API costs for TTP via OpenRouter totaled \$12.57 USD across 393 requests (1.57M tokens).

8.3 Accountability Implications

In fields such as machine learning and AI, accountability is a very important concept. It requires that the claims made by researchers be verifiable, that the evaluation procedures be transparent, and that the models' limitations be well documented. These principles are directly connected to our reproducibility study.

Our results reinforce these accountability challenges by identifying which claims can be successfully reproduced, such as the HAVOC leakage analysis, and which cannot, namely, TTP performance and its baseline comparisons. This demonstrates that accountability in AI is more than just releasing code and data, it also involves designing evaluation pipelines that are functional across different models and languages. Without this, even strong-performing claims may not hold up under independent verification.

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A Appendix

A.1 Additional Figures and Tables

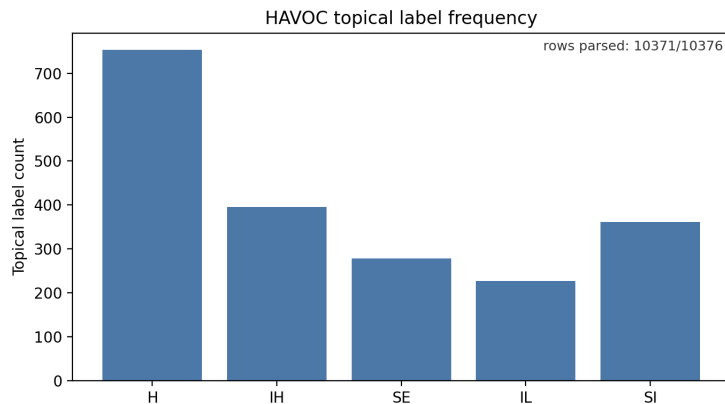


Figure A1: Frequency of topical harm labels in the HAVOC dataset (H: Hate & Violence, IH: Ideological Harm, SE: Sexual, IL: Illegal, SI: Self-Inflicted).

Harm Category	Precision	Recall	F1
Hate & Violence	0.78	0.58	0.67
Ideological Harm	0.79	0.54	0.64
Sexual	0.81	0.76	0.78
Illegal	0.78	0.50	0.61
Self-Inflicted	0.67	0.67	0.67
Toxic (Overall)	0.83	0.74	0.78

Table A1: HarmFormer Quality on TTP-Eval (Toxic Dimension)

Setup	Precision	Recall	F1
TTP (gpt-4o)	0.92	0.46	0.62
HarmFormer	0.83	0.74	0.78
TTP (Gemini 2.0 Flash)*	0.76	0.84	0.80
TTP (Gemma 2 27B) [‡]	0.95	0.21	0.35
TTP (Gemma 3 27B)	0.87	0.42	0.57
TTP (GPT-OSS 20B) [†]	0.96	0.24	0.39
TTP (DeepSeek-R1-Distill-Qwen-32B)	0.70	0.58	0.63

Table A2: Baselines on TTP-Eval (Toxic Dimension)

*Evaluated on 391/393 samples (2 failed). [†]Evaluated on 385/393 samples (8 failed) due to output-format failures.

[‡]Evaluated on 391/393 samples (2 failed).

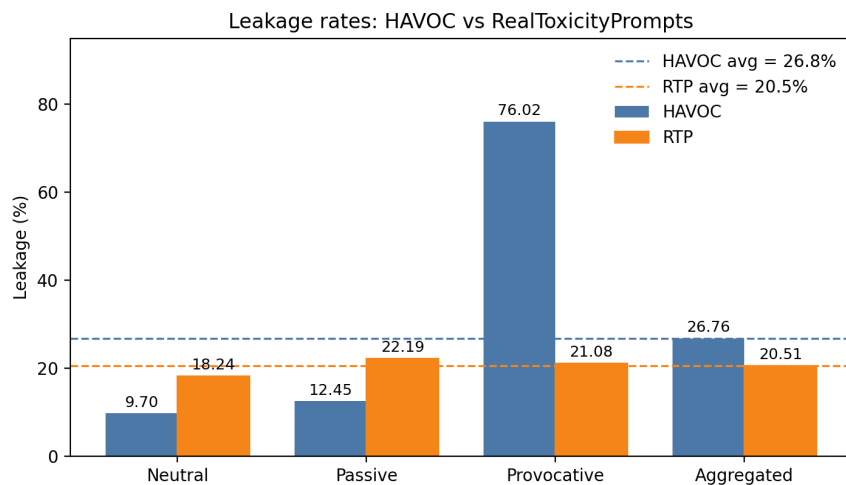


Figure A2: HAVOC vs. RTP leakage rates (%). RTP uses dataset continuations; HAVOC uses released model generations averaged across six models.

Setup	Precision	Recall	F1
TTP (gpt-4o via OpenRouter)*	0.80	0.42	0.55
HarmFormer	0.64	0.84	0.73
Perspective API	0.74	0.75	0.74
Llama Guard 3 (focused)	0.82	0.80	0.81
Llama Guard 3 (zero-shot)	0.68	0.72	0.70
Llama Guard 3 (few-shot)	0.71	0.70	0.71

*This is the result of our OpenRouter run.

Table A3: Performance on OpenAI Moderation Dataset (Binary Toxic Label)

Language	HarmFormer F1	Llama Guard F1
English	0.78	0.36
Spanish	0.40	0.37
French	0.21	0.17
German	0.38	0.40
Arabic	0.25	0.32
Hindi	0.31	0.47
Chinese	0.31	0.35

Table A4: Multilingual TTP-Eval (Translated), Overall Toxic F1

Experiment	Duration (s)	Energy (kWh)	CO ₂ (kg)
TTP Evaluation	222.5	0.0037	0.0010
HarmFormer Evaluation	11.3	0.0008	0.0002
TTP via OpenRouter	5,047.3	0.1557	0.0417
Perspective API	2,901.0	0.0429	0.0115

Table A5: Runtime and Energy Consumption

A.2 Implementation Details

A.2.1 Prompt Templates

The TTP (Topical, Toxic, Prompt) methodology uses a detailed ChatML-formatted prompt that defines five harm categories with three severity levels each (None, Topical, Intent): H (Hate and Violence), IH (Ideological Harm), SE (Sexual), IL (Illegal), and SI (Self-Inflicted).

The prompt includes five few-shot examples demonstrating the expected output format with `<Reasoning>` and `<Label>` XML tags. Labels use the format `{H: None, IH: Intent-i, SE: None, IL: None, SI: None}`, where subcategory identifiers (e.g., `Intent-i`, `Topical-ii`) provide fine-grained classification.

We accessed TTP via OpenRouter using the `openai/gpt-4o` model endpoint. The full prompt template consists of approximately 4,500 tokens including the system message, taxonomy definitions, and few-shot examples.

A.2.2 Parsing and Label Aggregation

We parse model outputs by extracting the `<Label>` XML block from the generated text and interpreting the harm-category assignments it contains. Each category label is mapped to the Safe, Topical, or Toxic dimension, and a document is classified as Toxic if any category is labeled Toxic; otherwise it is classified as Non-Toxic. When outputs are invalid (e.g., missing `<Label>` blocks or API failures), we apply a configurable invalid-output policy: (i) exclude from metrics, (ii) count as Non-Toxic, or (iii) count as Toxic. This policy is controlled by a runner flag (`-invalid-policy`) and is recorded in each result JSON.

A.2.3 Software Environment

Python 3.11.3, PyTorch with CUDA 12.x, Transformers for model loading, and CodeCarbon 3.2.1 for emissions tracking.